LinkedIn Salary: Curating and Evaluating a Text–Tabular Regression Benchmark

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Abstract

Job advertisements combine free-text descriptions of roles and requirements with structured employment and company information. We curate a dataset of 15,000 LinkedIn postings for predicting the natural logarithm of annualized advertised salary from two text fields and twelve structured features. Preparation includes masking recognized monetary expressions, integrating metadata, and deduplicating cleaned and truncated descriptions. Using the native MulTaBench curation pipeline, we evaluate five tabular learners under structured-only, text-only, joint frozen, and joint target-aware representation (TAR) conditions, comprising 100 learner–condition evaluations across five random holdout repetitions. Joint Frozen improves mean R^2 by 0.0234–0.0356 over the stronger unimodal baseline, while TAR yields additional gains of 0.0234–0.0668. Both Joint Signal and TAR Gain are positive for all five learners, satisfying the two curation criteria. These findings support complementary predictive information under the evaluated representations, while salary-disclosure selection, imperfect masking, and snapshot-based metadata limit broader generalization.

1 Introduction

Job postings describe responsibilities, qualifications, and seniority in free text, while also recording employment type, location, industry, and company characteristics. Predicting advertised salary therefore offers a setting for examining whether textual and structured information provide complementary predictive signals. However, the presence of both modalities does not guarantee that combining them improves prediction. Text descriptions may already encode information available in structured features, while explicit salary expressions could reduce the task to extracting the prediction target.

MulTaBench [1] addresses this distinction through two dataset curation criteria: joint prediction must outperform both unimodal baselines, and target-aware representations must outperform frozen embeddings. Each criterion must hold for at least three of the five tabular learners, counted independently. We apply this framework to a curated dataset of 15,000 LinkedIn job postings. Our contribution is a reproducible data curation pipeline with salary-leakage mitigation, together with an empirical characterization of the dataset across five learners and four experimental conditions. The objective is to determine whether the dataset satisfies both criteria and quantify the contributions of structured information and target-aware adaptation. The target represents advertised salary in a collected snapshot, rather than actual employee compensation.

2 Related Work

MulTaBench [1] introduces a benchmark curation framework emphasizing complementary predictive signals across modalities and improvements from target-aware representations. Relatedly, TabSTAR [2] explores adapting text representations for tabular prediction through a dedicated model archi-

tecture. Our study uses MulTaBench to assess dataset suitability and the benefit of task-aware text representations.

To evaluate the dataset across diverse modeling approaches, we use three families of tabular learners: parameter-efficient MLP ensembling with TabM [3], gradient-boosted decision trees with CatBoost [4] and LightGBM [5], and pretrained tabular foundation models TabPFN-v2 [6] and TabPFN-2.5 [7].

Text representations are generated using E5 [8], a family of contrastively pretrained text encoders. Target-aware adaptation uses LoRA [9], which introduces trainable low-rank updates while keeping the pretrained model weights frozen. Our evaluation follows MulTaBench's E5-based representation and adaptation procedures.

3 Dataset and Curation

Source and target. We use arshkon's LinkedIn Job Postings (2023–2024) [10], distributed under CC BY-SA 4.0. The local source contains 123,849 postings and linked company, industry, skill, and benefit tables. The target is $y = \ln(s)$, where s is the source-provided annualized USD normalized salary. Our preparation uses this existing field rather than implementing a new annualization procedure.

Selection and cleaning. We retain 36,058 postings with a nonmissing USD salary, then restrict salaries to USD 15,000–1,000,000, leaving 35,567. In titles and descriptions, regular expressions replace recognized monetary expressions with `[amount]` and matching contact details with `[email]` or `[phone]`. Whitespace is normalized, and descriptions are limited to their first 500 whitespace-separated words. Removing one empty description and 2,673 repeated cleaned and truncated descriptions leaves 32,893 eligible postings. We draw 15,000 without replacement using seed 42. No explicit country or language filter is applied.

Features. The text fields are title and description. Six categorical inputs describe experience level, work type, state, application type, company industry, and job industry. Six numerical inputs describe the remote flag, company-size code, employee count, follower count, and numbers of linked skill and benefit records. State is extracted from a terminal two-letter location code; industry fields use the first available mapping. Missing remote flags and absent skill/benefit records become zero; remaining missing values are retained for native learner preprocessing. State and industry categories with fewer than twenty sampled occurrences are merged into Other before evaluation.

Salary fields, pay periods, identifiers, company-name columns, URLs, timestamps, and engagement counts are excluded from predictors. Company names can nevertheless remain in free text. Company-size codes are distinct from employee headcounts, and employee/follower metadata use the latest available source records.

Dataset properties. Annualized salary has median USD 83,750 and a fifth–ninety-fifth percentile range of USD 35,360–200,000. Median title and description lengths are three and 483 words. All 15,000 cleaned descriptions are unique. The masking token `[amount]` appears in 9,504 descriptions (63.36%); this measures the occurrence of recognized expressions, not the recall of salary masking. Missingness remains substantial for experience level (22.43%) and state (18.04%).

4 Experimental Setup and Results

4.1 Native Evaluation Protocol

We use the official MulTaBench implementation from 26 July 2026 (commit 599bd6a) to evaluate TabM, CatBoost, LightGBM, TabPFN-v2, and TabPFN-v2.5. LinkedIn Salary is integrated through the native dataset registry, local-file mapping, and curation annotation, without modifying the benchmark's learning, preprocessing, splitting, or metric logic.

Each learner is evaluated under four conditions: Structured (S), using only the twelve structured features; Text Frozen (T), using frozen text representations; Joint Frozen (JF), combining structured and frozen text features; and Joint TAR (JT), combining structured features with target-aware text representations.

Repeated evaluation. The native protocol uses five random repetitions (seeds 42–46). Each repetition selects 11,500 postings from the 15,000-row dataset, with 10,000 used for outer training and 1,500 for testing. These are repeated random holdouts rather than disjoint cross-validation folds, and corresponding learners and conditions use the same repetition.

TabM, CatBoost, and LightGBM reserve 1,000 outer-training observations for learner validation and fit preprocessing on the remaining 9,000. Both TabPFN learners use all 10,000 outer-training observations.

Representations and adaptation. Frozen and target-aware text representations use intfloat/e5-small-v2 with column-prefixed inputs and a 512-token limit. Training-fitted PCA reduces normalized text embeddings to 30 dimensions per field, yielding 60 text-derived features.

For TAR, regression targets are discretized into up to 20 quantile bins, and E5 is adapted using LoRA through an auxiliary classification objective. Adaptation uses 8,100 training and 900 validation rows for TabM, CatBoost, and LightGBM, and 9,000/1,000 for the TabPFN learners. Each posting provides separate title and description examples. A separate encoder is adapted for each learner and repetition (25 fits overall), with outer-test data excluded from fitting and early stopping.

LoRA uses rank 16 on the final three layers, learning rate 10^{-4}, weight decay 0.001, and a maximum of 50 epochs with early-stopping patience three. TAR training uses physical batch size 256, without gradient accumulation, in float32. Experiments use NVIDIA A40 GPUs, with one GPU assigned to each job.

Metrics and acceptance criteria. Performance is measured using native R^2 on log salary, with mean and sample standard deviation reported across five repetitions in the final results. Native MSE and RMSE are retained as supplementary metrics. For each learner, Joint Signal (Δ_{Joint}) is the mean JF score minus the stronger unimodal mean, while TAR Gain (Δ_{TAR}) is the mean JT score minus the mean JF score. Both differences use unrounded scores. Each criterion must be positive for at least three of five learners, counted independently. Variation across repeated holdouts is descriptive and is not interpreted as statistical significance.

4.2 Primary Native Results

All 100 expected native evaluations completed successfully. Table 1 summarizes mean R^2 and sample standard deviation across five random holdout repetitions.

Learner	S	T	JF	JT	Δ_{Joint}	Δ_{TAR}
TabM	0.4521 $\pm$ 0.0173	0.6858 $\pm$ 0.0230	0.7122 $\pm$ 0.0169	**0.7461 $\pm$ 0.0126**	+0.0264	+0.0338
CatBoost	0.4842 $\pm$ 0.0107	0.6544 $\pm$ 0.0201	0.6870 $\pm$ 0.0208	**0.7367 $\pm$ 0.0184**	+0.0326	+0.0497
LightGBM	0.4703 $\pm$ 0.0157	0.6268 $\pm$ 0.0192	0.6624 $\pm$ 0.0171	**0.7292 $\pm$ 0.0170**	+0.0356	+0.0668
TabPFN-v2	0.4870 $\pm$ 0.0204	0.6674 $\pm$ 0.0229	0.6908 $\pm$ 0.0200	**0.7320 $\pm$ 0.0183**	+0.0234	+0.0412
TabPFN-v2.5	0.4841 $\pm$ 0.0271	0.6926 $\pm$ 0.0223	0.7171 $\pm$ 0.0193	**0.7405 $\pm$ 0.0162**	+0.0245	+0.0234

Table 1: Native mean R^2 with $\pm$ sample standard deviation beneath each mean, across five repetitions. S: Structured; T: Text Frozen; JF: Joint Frozen; JT: Joint TAR. Δ_{Joint} compares JF with the stronger unimodal mean, while Δ_{TAR} compares JT with JF. Gains use unrounded means.

Text Frozen outperforms Structured for all five learners. Combining structured features with frozen text representations yields joint gains of 0.0234–0.0356 R^2, averaging 0.0285 across learners. Target-aware adaptation provides additional gains of 0.0234–0.0668, averaging 0.0430. These improvements indicate that structured metadata contributes information beyond frozen text representations, while target-aware adaptation provides further predictive value.

Both Joint Signal and TAR Gain are positive for all five learners, satisfying the two curation criteria and exceeding the course requirement of at least three learners per criterion. TabM achieves the

highest mean Joint TAR performance ($R^2 = 0.7461$), while LightGBM exhibits the largest joint and TAR gains (0.0356 and 0.0668, respectively).

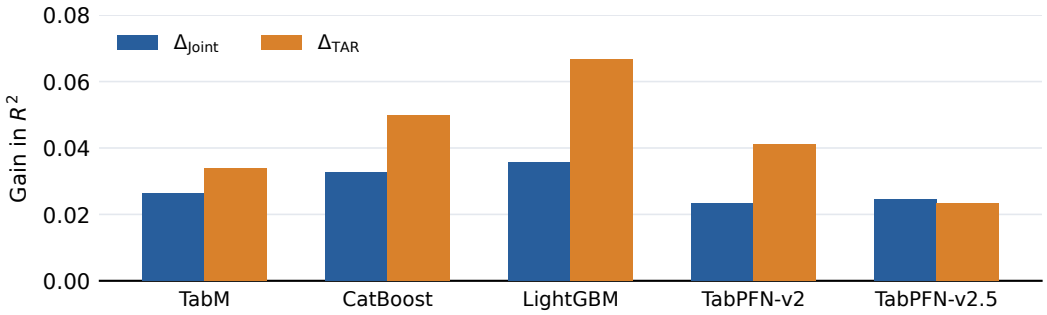


Figure 1: Native Joint Signal (Δ_{Joint}) and TAR (Δ_{TAR}) gains across five repetitions. Δ_{Joint} is mean JF minus the stronger unimodal mean; Δ_{TAR} is mean JT minus mean JF. Bars represent differences between condition means, not confidence intervals.

These results characterize performance under the evaluated native protocol and do not constitute a statistical significance test.

4.3 Earlier Evaluation: Supporting Evidence

Before the primary native evaluation, we completed a physical-batch-256 experiment using a project harness around official MulTaBench components from the 21 October 2026 snapshot. It used the same curated dataset and three-fold cross-validation repeated twice, producing 120 evaluation records across five learners and four conditions. All five learners achieved positive mean Joint Signal gains (0.0254–0.0373 R^2) and TAR gains (0.0217–0.0738 R^2). Full results are reported in Appendix A.

These findings provide supporting evidence for the dataset's predictive properties but do not constitute a direct replication of the native evaluation. The evaluations differ in source versions, splitting procedures, preprocessing, and TAR encoder fitting. Performance differences therefore cannot be attributed to the execution framework alone.

5 Conclusion, Limitations, and Future Work

We curated a 15,000-row salary-prediction dataset and evaluated it through the native MulTaBench framework. Based on mean performance across five random holdout repetitions, both Joint Signal and TAR Gain are positive for all five learners, satisfying the course's curation criteria. An earlier harness evaluation showed the same qualitative outcome, although the protocols differ.

Several limitations qualify these findings. Filtering for reported USD salaries within specified bounds introduces selection bias, limiting generalization to all job advertisements or actual employee compensation. Pattern-based masking may miss salary cues or remove unrelated numerical information; its precision and recall were not measured. Exact-description deduplication does not eliminate near-duplicates or employer overlap.

Company metadata are snapshot-based: 11,631 of 14,790 linked postings use company-count records timestamped after the advertisement, potentially introducing information unavailable at posting time. The effect of this temporal mismatch was not evaluated. Row-based splitting and dataset-wide rare-category consolidation further limit conclusions about prospective or unseen-employer performance.

Finally, repeated holdouts overlap, so the reported variability is descriptive rather than evidence of statistical significance. Future work should evaluate employer-grouped and chronological splits, use time-consistent metadata, audit residual salary cues and near-duplicates, and examine sensitivity to text truncation and representation choices.

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A Earlier Harness Evaluation

Learner	S	T	JF	JT	Δ_{Joint}	Δ_{TAR}
TabM	0.4677 ± 0.0105	0.6901 ± 0.0073	0.7194 ± 0.0072	0.7490 ± 0.0100	$+0.029259$	$+0.029613$
CatBoost	0.4923 ± 0.0095	0.6545 ± 0.0073	0.6918 ± 0.0075	0.7419 ± 0.0110	$+0.037300$	$+0.050094$
LightGBM	0.4807 ± 0.0108	0.6249 ± 0.0080	0.6617 ± 0.0053	0.7354 ± 0.0090	$+0.036738$	$+0.073765$
TabPFN-v2	0.4970 ± 0.0106	0.6707 ± 0.0053	0.6975 ± 0.0051	0.7358 ± 0.0090	$+0.026866$	$+0.038289$
TabPFN-v2.5	0.5045 ± 0.0105	0.6975 ± 0.0064	0.7229 ± 0.0066	0.7446 ± 0.0100	$+0.025403$	$+0.021713$

Table 2: Completed physical-batch-256 harness evaluation using the 21 October source snapshot. Scores are mean R^2 with $\pm$ sample standard deviation beneath each mean, across six cross-validation splits. S: Structured; T: Text Frozen; JF: Joint Frozen; JT: Joint TAR. $\Delta_{\mathrm{Joint}} = \overline{R^2}_{\mathrm{JF}} - \max(\overline{R^2}_{\mathrm{S}}, \overline{R^2}_{\mathrm{T}})$; $\Delta_{\mathrm{TAR}} = \overline{R^2}_{\mathrm{JT}} - \overline{R^2}_{\mathrm{JF}}$. Deltas use unrounded means and are shown to six decimal places. These are supporting historical results, not the primary native results.